

Using Android for the Enterprise



This article takes an in-depth look at Android devices in relation to their use in the enterprise. The author details the features, form factors and strategies for enterprise mobile use. He also covers aspects like customisation, applications and management of both Android devices and apps.

The Android mobile operating system by Google has been primarily designed for devices such as touchscreen mobiles and tablets, with specialised versions for TV (Android TV), cars (Android Auto) and wearable devices (Android Wear). It is one of the leading mobile operating systems across the world. Its source code has been released under an open source licence. The application store, Google Play, provides the platform to upload/download Android mobile apps. Features supported by the Android platform are:

- Bar code reader
- OCR (optical character recognition) integration
- NFC (near field communication) wallets
- LBS (location based services) navigation, directions
- Map view, Geo tracking/tagging/fencing
- Push notification – GCM (Google Cloud Messaging), C2DM (Cloud-to-Device Messaging)
- Data sharing across applications using a content provider
- Relational database support - SQLite

By using Android supported features, distinctive applications can be implemented, such as:

- Interactive UI based charting, reporting applications
- Relational database-intensive applications, high performance SQLite based applications
- Native widget development
- UI customisation for rich UI and animation based applications with features like ‘wise view’, drag-and-drop, list reordering, etc
- Native application development using NDK (native development kit)

Device form factors

Evolving hardware capabilities are the major cause for rapidly evolving mobile devices, and their availability in diverse form factors. Android devices are available on the following form factors:

- *Smartphones:* These mobile computing devices generally